

Sat, Sep 12 at 2:11 PM

hey i got selected for ai research by kohler, now there is a case study and interview i have to submit the case study on 20th and interview is on 22.

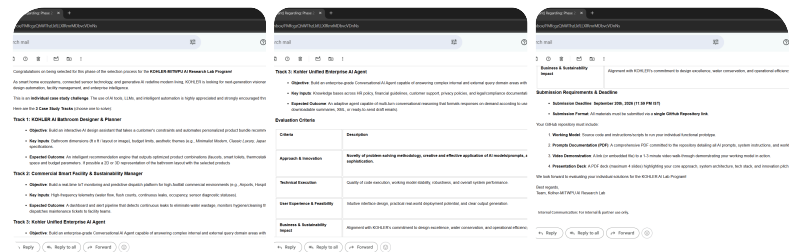
That's great news—congratulations! 🎉 Getting selected for an **AI Research** role at Kohler is a strong opportunity.

Since your **case study is due on September 20** and the **interview is on September 22**, we should treat this as two connected tasks:

1. **Case study:** build a strong, research-oriented solution that shows your AI/ML thinking, not just implementation.
2. **Interview:** prepare you to defend every part of the case study, plus likely AI/ML, research, and problem-solving questions.

We have **8 days until the submission** and **10 days until the interview**, so there's enough time to make the case study polished rather than rushing it.

Send me the **Kohler case study/problem statement** and **all the instructions they gave you** (screenshots, PDF, email, rubric, dataset, submission format, etc.). I'll help you break it down, decide the strongest approach, develop the solution, and then prepare the interview questions directly around it.



Yes — now we have the **actual selection brief**, and this changes how I would approach the preparation.

You are not being asked to merely build an AI demo. The judging heavily favors **how creatively and intelligently you solve the problem**.

What KOHLER is actually evaluating

Criterion	Weight	What this means for us
Approach & Innovation	45%	This is the biggest one. Novel architecture, clever AI usage, strong prompting, optimization, agents, multimodal AI, etc.
Technical Execution	25%	It has to actually work and be reasonably robust.
UX & Feasibility	20%	The demo should feel like something KOHLER could genuinely use.
Business & Sustainability	10%	Connect it to water conservation, efficiency, design, operational value, etc.

So our goal should **not** be:

"I made an LLM application."

It should be:

"I designed an intelligent system that solves a real KOHLER problem using AI in a way that would be difficult to solve with a simple chatbot or conventional software."

My recommendation: Track 1

KOHLER AI Bathroom Designer & Planner

I would personally choose Track 1 for you.

The reason is that it gives us opportunities to combine several AI capabilities in one system:

Multimodal AI + LLM + recommendation engine + constraint optimization + product knowledge + spatial reasoning + personalized UX

That directly attacks the **45% innovation** category.

The basic assignment is:

User gives bathroom dimensions/layout/image + budget + aesthetic preference → system recommends a physically compatible bathroom/product bundle.

But we should **not stop there**.

A weak solution would look like:

User:

"My bathroom is 8 × 6 ft and budget is ₹2 lakh."

LLM:

"Here are 5 Kohler products..."

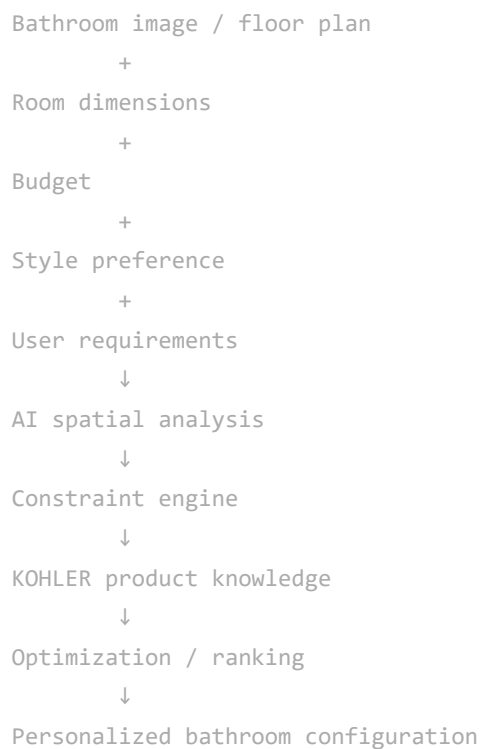
That would be easy to build and probably not very impressive.

The solution I would build

I would turn it into something like:

KOHLER AI Space-to-Bathroom Planner

A multimodal AI system that takes:



And the output isn't just a product list.

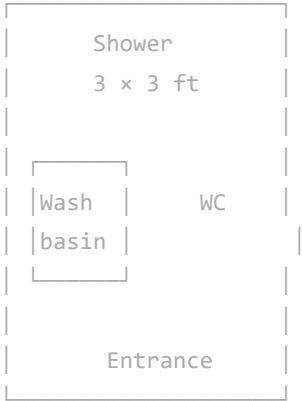
It gives:

1. Spatial feasibility

For example:

Bathroom: 8 × 6 ft

The system determines approximate usable zones:



It checks whether the selected products can physically coexist.

2. Budget optimization

Suppose:

Budget = ₹1,50,000

Instead of simply recommending expensive products, the system can optimize:

Product compatibility
+
Style compatibility
+
Space compatibility
+
Budget
+
User preferences

to generate the **best overall configuration**.

For example:

Component	Product	Price	Reason
Toilet	Product A	₹XX,XXX	Space efficient
Basin	Product B	₹XX,XXX	Matches style
Faucet	Product C	₹XX,XXX	Compatible
Shower	Product D	₹XX,XXX	Budget optimized

Then:

Total: ₹1,47,500

Here's where we can make it significantly better

I would add a two-stage AI architecture.

Stage 1 — AI understands the user

LLM / Vision model extracts:

```
{
  "room_length": 8,
  "room_width": 6,
  "style": "Japanese Zen",
  "budget": 150000,
  "requirements": [
    "smart toilet",
    "rain shower",
    "vanity"
  ]
}
```

Stage 2 — Deterministic optimization

This is important.

We shouldn't allow the LLM to randomly decide whether something fits.

Instead:

```
LLM
↓
Structured requirements
↓
Product database
↓
Constraint filter
↓
Optimization algorithm
↓
LLM explanation
```

So:

LLM = reasoning/interface

Constraint engine = factual decision maker

This makes the architecture much more defensible in the interview.

We can even introduce a scoring function

For every possible bathroom configuration:

$$Score = w_1SpaceFit + w_2BudgetFit + w_3StyleMatch + w_4FeatureMatch + w_5Sust$$



For example:

Space Fit	30%
Budget	25%
Style	20%
Requirements	15%
Sustainability	10%

Then the recommendation engine ranks configurations.

That immediately makes the project feel more like **AI research / intelligent optimization**, rather than simply an LLM wrapper.

An especially strong innovation

I would add a feature called something like:

"Why this design?"

The system explains its recommendation.

For example:

Why this configuration?

This vanity was selected because its width leaves approximately 24 inches of clearance from the entrance path while keeping the total configuration within your ₹1.5 lakh budget.

That gives us **explainability**.

And during the interview, you can say:

"I deliberately separated generative reasoning from deterministic constraints because an LLM should not be trusted to make physical compatibility decisions."

That is a **very good research-oriented answer**.

We could also make the system iterative

Instead of a one-shot recommendation:

User

I don't like this bathroom. Make it more luxurious.

AI

Updates:

Style weight ↑
Budget weight ↓
Premium products ↑

Then:

What if I increase my budget to ₹2 lakh?

The system recalculates the configuration.

Then:

Keep the toilet but make the shower more premium.

Only the appropriate part of the configuration changes.

That demonstrates **multi-turn agentic reasoning**.

Why I don't immediately recommend Track 3

Track 3 is:

KOHLER Unified Enterprise AI Agent

It's absolutely possible to make a strong project there, but enterprise RAG agents are becoming a very common AI project pattern.

You could build:

```
Documents
  ↓
RAG
  ↓
LLM
  ↓
Agent
  ↓
JSON / Excel / Email
```

That's useful, but we'd need to work much harder to make the **innovation** stand out.

Track 1 gives us a much more distinctive demonstration.

Track 2 is actually very interesting too

Track 2 could become highly technical:

```
IoT telemetry
  ↓
Time-series processing
  ↓
Anomaly detection
  ↓
Leak detection
  ↓
Usage prediction
  ↓
Maintenance prioritization
  ↓
Automatic dispatch
```

You could introduce:

- anomaly detection
- forecasting
- predictive maintenance
- dynamic alert thresholds
- optimization
- sustainability metrics

Technically, this could be extremely strong.

The problem is that we'd need to create a convincing **synthetic IoT dataset/simulation** for the prototype.

So my ranking for your case study would be:

Track 1 — Best overall

AI Bathroom Designer & Planner

Track 2 — Best for pure technical/ML depth

Smart Facility & Sustainability Manager

Track 3 — Best if you want an LLM/agent-heavy project

Unified Enterprise AI Agent

What we need to submit

Your GitHub repository has four important deliverables:

```
KOHLER Case Study
|
├── 1. Working Model
|   ├── source code
|   ├── setup instructions
|   └── run instructions
|
├── 2. Prompts Documentation
|   └── PDF containing prompts,
|       system instructions,
|       workflows
|
├── 3. Video Demonstration
|   └── 1-3 minute demo
|
└── 4. Presentation Deck
    └── maximum 4 slides
```

And this is important:

The 4-slide limit actually works in our favor.

We can design the presentation as:

Slide 1 — Problem + Innovation

Slide 2 — AI/System Architecture

Slide 3 — Working Demo + Technical Execution

Slide 4 — Business Impact + Future Scalability

Our timeline

You have until **September 20 at 11:59 PM IST**.

I would work on it in this order:

Sept 12–13

Finalize problem interpretation + architecture + innovation.

Sept 14–16

Build the working prototype.

Sept 17

Improve UI + recommendation quality + edge cases.

Sept 18

Create prompts documentation + architecture diagrams + test cases.

Sept 19

Record demo + make 4-slide presentation.

Sept 20

Final GitHub cleanup, testing and submission.

Then:

Sept 21

Interview preparation based specifically on your implementation.

Sept 22

Interview.

And I don't think we should start coding immediately.

First, we should design the solution that we're going to build, because the architecture and innovation are worth **45% of the evaluation**.

I suggest we next build the complete **Track 1 solution blueprint** together: exact features, AI models, architecture, product-data structure, recommendation algorithm, tech stack, UI, and what we will actually implement versus simulate.

1. Python for AI/ML
2. Deep Learning fundamentals
3. Neural Networks
4. Transformers
5. LLMs (Large Language Models)
6. Prompt Engineering
7. System Prompts & Instruction Hierarchy
8. Tokens & Context Windows
9. Embeddings
10. Semantic Similarity
11. Vector Databases
12. Vector Search
13. RAG (Retrieval-Augmented Generation)
14. Document Loading & Parsing
15. Chunking & Chunking Strategies
16. Metadata & Metadata Filtering
17. Retrieval & Reranking
18. RAG Evaluation
19. Hallucination & Grounding
20. LLM Structured Outputs
21. Function Calling / Tool Calling
22. APIs
23. AI Agents
24. Agent Architecture
25. Agent Planning
26. Reasoning & Task Decomposition
27. Agent Memory
28. Short-Term vs Long-Term Memory
29. Agent Loops (Think → Act → Observe)
30. Single-Agent vs Multi-Agent Systems
31. Agent Routing
32. Tool Selection
33. Workflow Orchestration

34. LangChain
35. LangGraph
36. State Management in Agents
37. Human-in-the-Loop
38. Guardrails
39. Prompt Injection & Jailbreaks
40. Access Control / Permission-Aware AI
41. Enterprise RAG
42. Knowledge Bases
43. Output Parsers
44. JSON Schema
45. Dynamic Output Generation
46. Email / Excel / File Generation through Tools
47. Agent Evaluation
48. Observability & Logging
49. Latency & Cost Optimization
50. Production AI / Deployment Basics

ok

👍 Great. We'll build your **KOHLER Track 3 project** around those concepts and keep the learning focused on what you actually need to understand the system.

Sat, Sep 12 at 7:47 PM

can we start research and everything in a new chat?

Yes, absolutely. We can continue the **KOHLER Track 3 project** in a new chat and treat that chat as the dedicated workspace for the case study.

In the new chat, just write something like:

"Continue my KOHLER AI Research Lab case study. We selected Track 3 — Unified Enterprise AI Agent. Let's start the research and project planning."

I'll have the relevant context from this conversation, including your goal of becoming an **Agentic AI Engineer**, so we can start directly with:

problem research → existing solutions → gaps → innovation idea → system architecture → technologies → implementation plan → case study → demo → presentation → interview preparation.

For the new chat, I recommend we start with researching the actual Track 3 problem and KOHLER's requirements before deciding the final architecture.